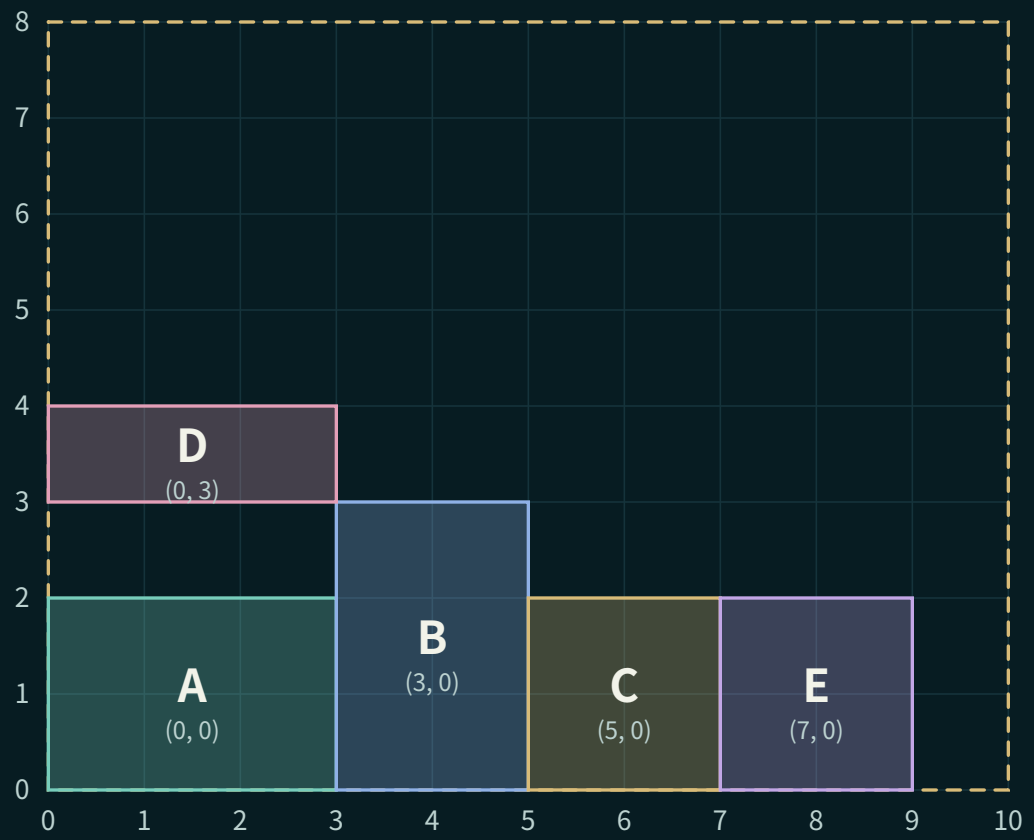
Two Rows of Letters

Sequence-Pair · Floorplanning lesson 6

Floorplanning · Lesson 6 · example and convention from the instructor's learn_floorplanning (CC BY 4.0)

The problem

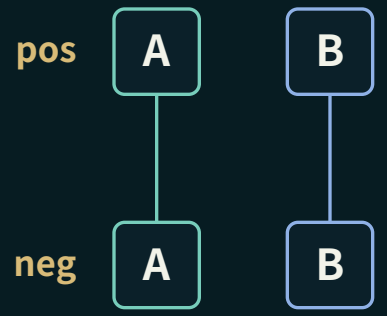
- Input: five rectangles (fixed size, no rotation), a 10 × 8 outline, permutations pos and neg
- Output: lower-left (x, y) of every block; graphs H and V; bounding box and metrics
- Convention: origin lower-left, y up (instructor's common/pack.py)
- Why not search coordinates: almost every combination overlaps

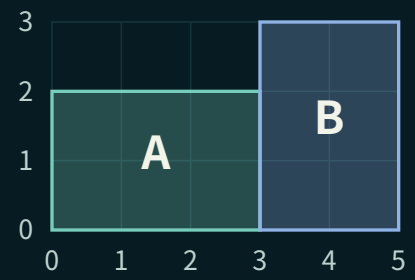


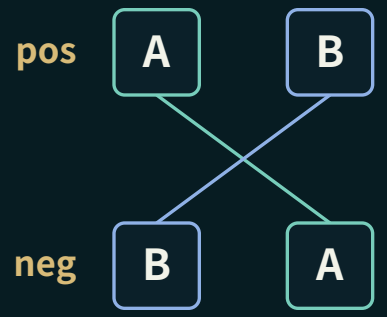
Block area 23, outline area 80: enough area does not mean it fits.

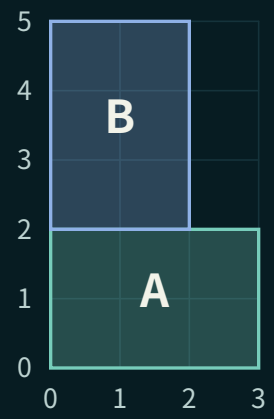
Two permutations, four relations

- First in both → left of: x[b] ⩾ x[a] + w[a]
- First in pos, second in neg → below: y[b] ⩾ y[a] + h[a]
- Rule of thumb: same order → left/right; opposite → below/above
- The other convention says 'above': same as swapping pos and neg



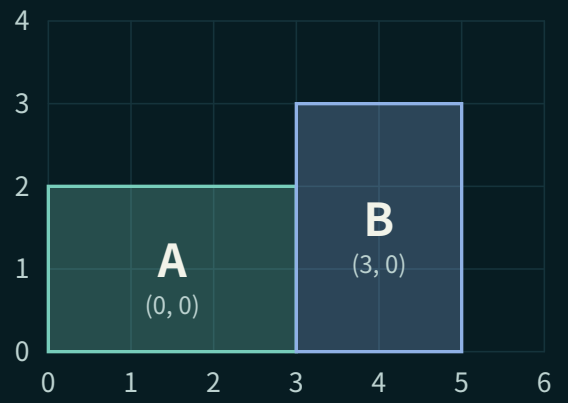


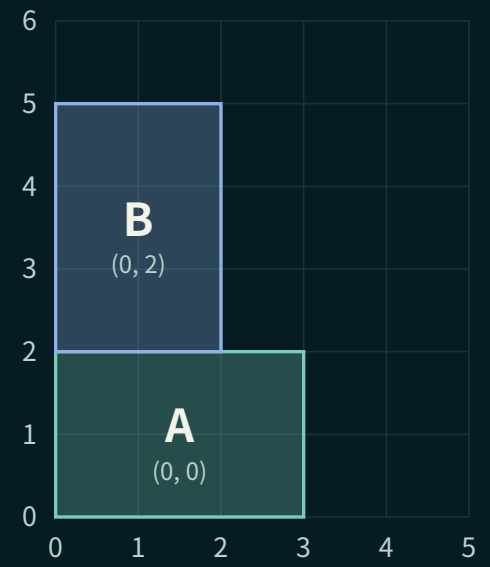




The smallest example: two blocks

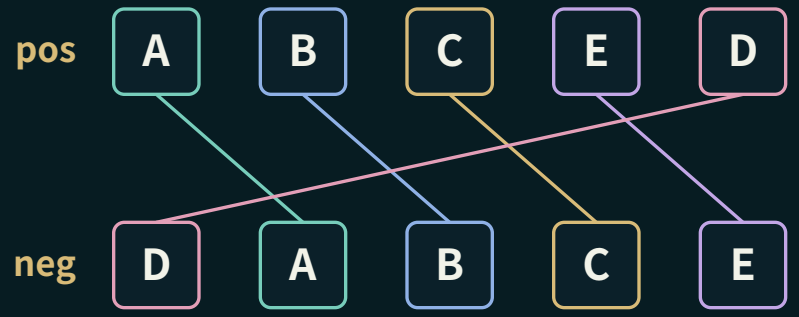
- A 3 × 2, B 2 × 3
- pos = A B, neg = A B: A left of B, B starts at x = 3
- pos = A B, neg = B A: A below B, B starts at y = 2
- Every pair has exactly one relation, never none, never two





The ten relations of the instructor's pair

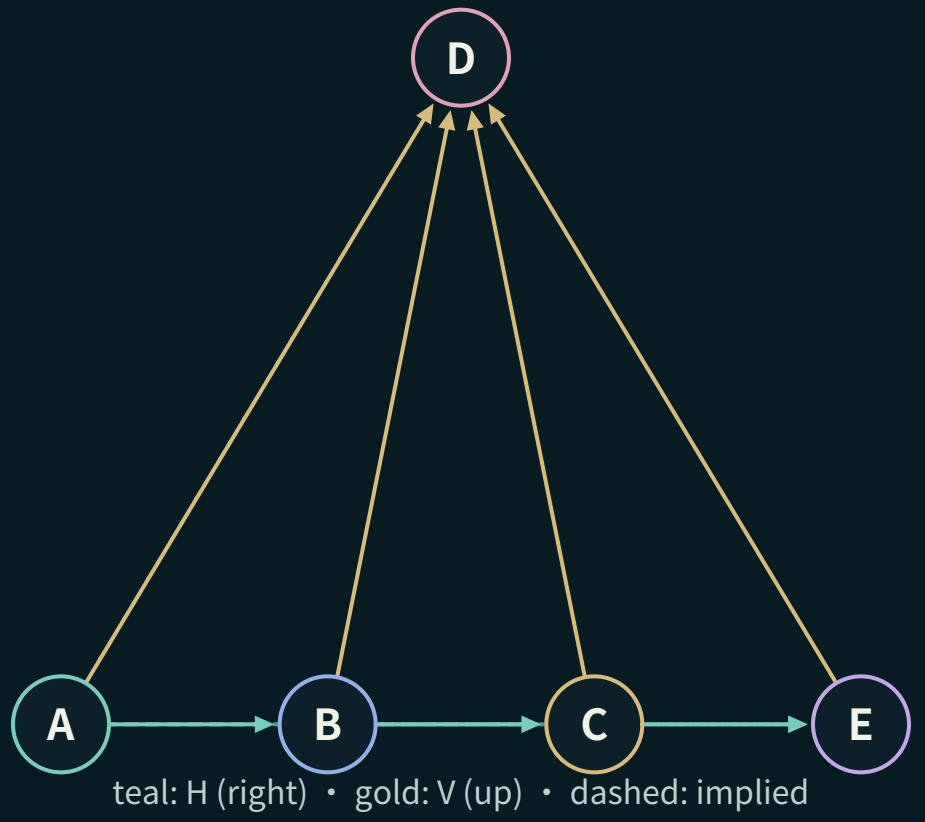
- pos = A B C E D, neg = D A B C E
- 5 blocks → 5 × 4 / 2 = 10 pairs
- 6 left/right: A, B, C, E in a row
- 4 below/above: D above the other four



Pair	Relation	Edge
A, B	A left of B	H: A → B
A, C	A left of C	H: A → C
A, E	A left of E	H: A → E
A, D	A below D	V: A → D
B, C	B left of C	H: B → C
B, E	B left of E	H: B → E
B, D	B below D	V: B → D
C, E	C left of E	H: C → E
C, D	C below D	V: C → D
E, D	E below D	V: E → D

Constraint graphs H and V

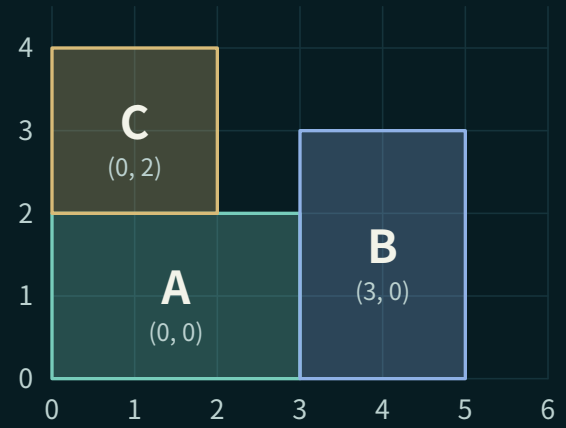
- H: i left of j → edge i → j, x[j] ⩾ x[i] + w[i]
- V: i below j → edge i → j, y[j] ⩾ y[i] + h[i]
- Each pair in one graph only: 6 in H, 4 in V
- Every edge points forward in pos → no cycle



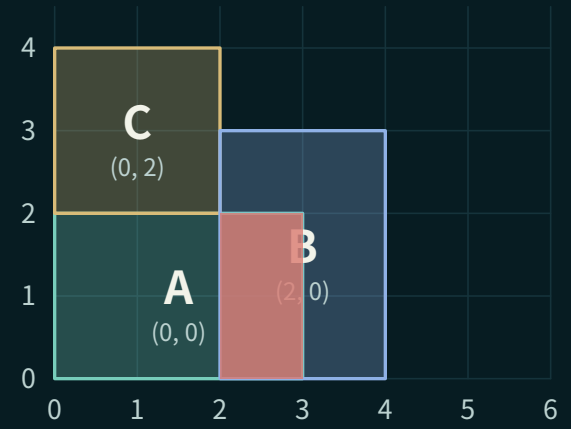
A → E is implied by A → B → C → E; drawn dashed, still used.

Why add the predecessor's width, why the maximum

- Edge i → j: j's left side stays right of i's right side x[i] + w[i]
- Every incoming edge is a lower bound; all must hold
- Smallest value meeting all bounds = the largest bound
- Three blocks: the minimum 2 makes B overlap A



maximum x[B] = 3: no overlap



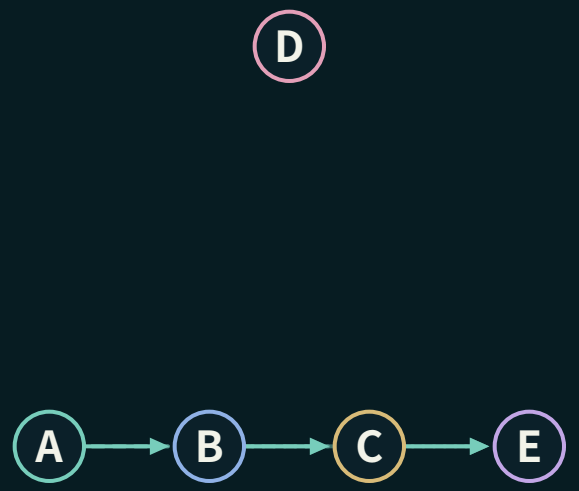
minimum x[B] = 2: overlap

Computing x in pos order

- x[A] = 0, x[B] = 3, x[C] = 5
- x[E] = max(0 + 3, 3 + 2, 5 + 2) = 7
- x[D] = 0: D has no horizontal predecessor
- This is the longest path in H

Pred.	Bound	Value
A	x[A] + w[A] = 0 + 3	3
B	x[B] + w[B] = 3 + 2	5
C	x[C] + w[C] = 5 + 2	7

maximum: x[E] = 7

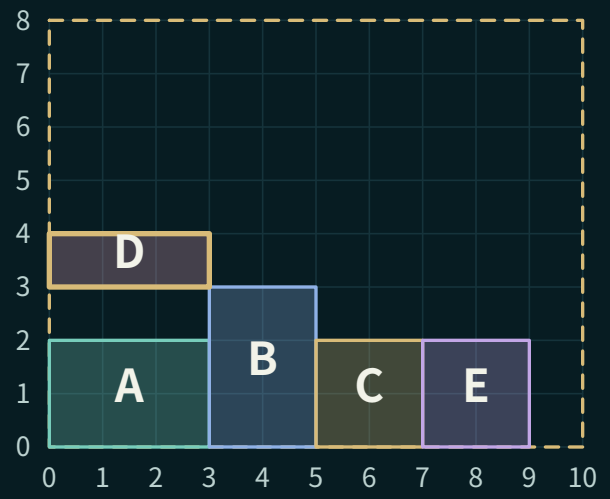


Then y

- A, B, C, E have no vertical predecessor: y = 0
- y[D] = max(2, 3, 2, 2) = 3: the tallest, B, decides
- The instructor's GOLDEN_PACK (D at y = 2) is a separate illustration, not this pair's output

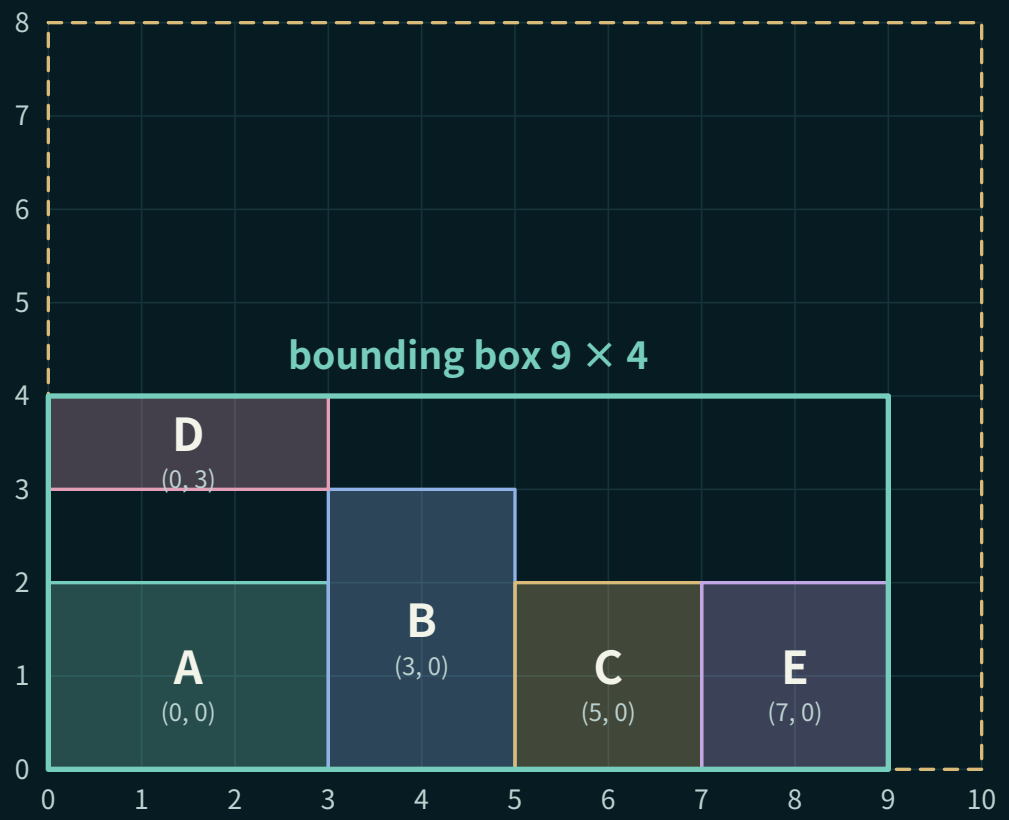
Pred.	Bound	Value
A	y[A] + h[A] = 0 + 2	2
B	y[B] + h[B] = 0 + 3	3
C	y[C] + h[C] = 0 + 2	2
E	y[E] + h[E] = 0 + 2	2

maximum: y[D] = 3



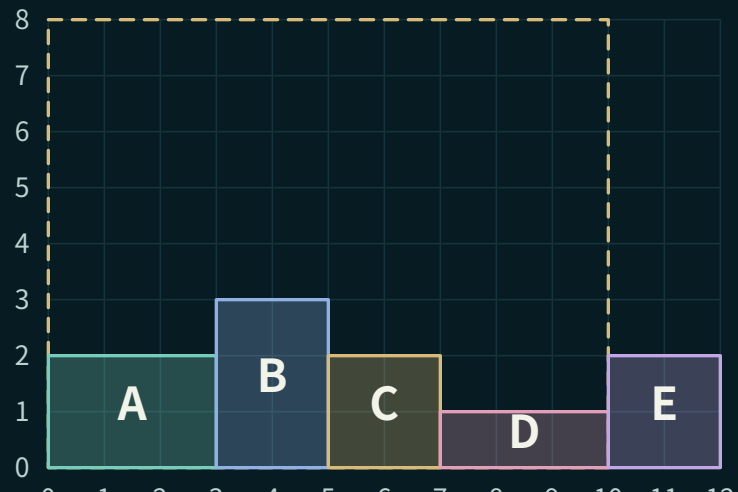
The result and four questions

- Complete: each block once
- No overlap: guaranteed by the pair
- Fits: box $9 \times 4 \leqslant 10 \times 8$
- Compact: 36, 13 unused, utilisation 63.89 %



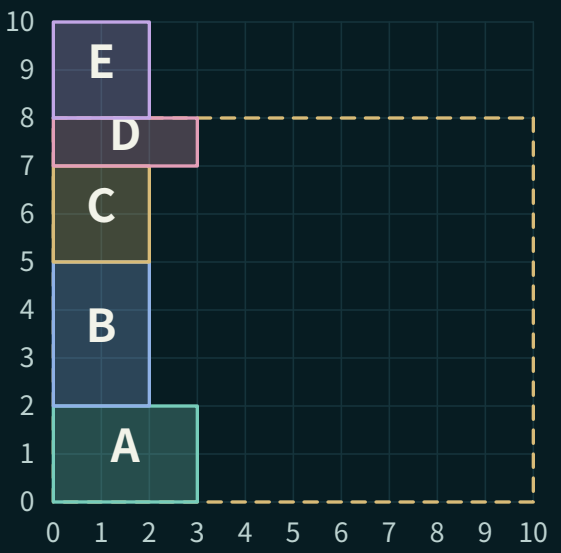
Valid ≠ fits; constants and variables

- pos = neg = A B C D E → 12 × 3, too wide
- neg reversed → 3 × 10, too tall
- 14,400 pairs: all overlap-free, 13,680 fit, 720 do not
- Outline unused 57, density 28.75 %: the same for every pair



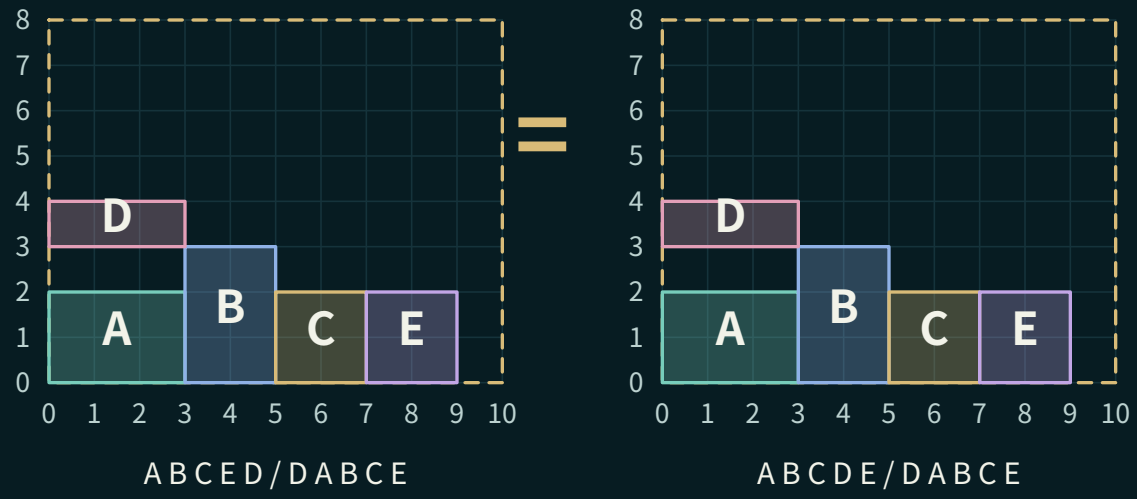
A B C D E / A B C D E → 12 × 3

A B C D E / E D C B A
→ 3 × 10



Changing a permutation

- Swap E and D in pos: the relation changes, the coordinates do not
- Swap D and A in neg: still a 9 × 4 box, a different packing
- 14,400 pairs give only 9,056 packings; the instructor's packing has 3 encodings
- Search (annealing) walks between these neighbours



Change	Box	Coords
pos A↔B	7 × 6	moved
pos B↔C	7 × 6	moved
pos C↔E	7 × 5	moved
pos E↔D	9 × 4	same
neg D↔A	9 × 4	moved
neg A↔B	7 × 6	moved

Pseudocode and cost

- Line 1: list every problem
- Lines 2–4: one edge per pair
- Lines 5–7: two DP passes in pos order
- Line 8: report four questions separately
- Cost $O(n^2)$; FAST-SP reaches $O(n \log n)$ (not implemented)

```
1  check pos, neg (every problem)
2  for each pair (i, j) in pos:
3    first in neg too → H edge
4    else → V edge
5  order ← pos
6  x[j] ← max(0, x[i] + w[i])
7  y[j] ← max(0, y[i] + h[i])
8  report four questions
```

Limits and next steps

A sequence pair writes relative positions as two permutations; coordinates come from longest paths.

Here: fixed sizes, no rotation; evaluation only, no search; no wires, timing or power.

Next: simulated annealing over sequence pairs. Source: Murata et al., IEEE TCAD 1996.